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Open Source Contributions

Mesa

2025 – Present

GPU drivers and shader compilers

- Investigate compiler and driver correctness issues across Mesa, from shader IR and code generation to Vulkan driver behavior.
- Contribute fixes across Intel's Jay and BRW shader compilers; diagnose failures with targeted Vulkan reproducers, Vulkan CTS, IR inspection, and sanitizer builds.
- Develop and test Mesa changes on Intel graphics (ANV) and Arm Mali-G610 (Panfrost) hardware.

Linux kernel

2025 – Present

Platform drivers and DRM

- Contribute focused fixes across platform/x86 and DRM drivers, including [hp-wmi](#), [i915](#) and [Panthor](#).
- Diagnose memory-safety, concurrency, undefined-behavior, and error-path issues with focused reproducers and sanitizer builds.

LLVM

2026 – Present

MC/ARM

- [Merged a fix](#) to preserve label-time ISA state for Thumb function symbols when a later `.type` directive marks the function.

Apache NuttX

2026

Arm64 SoC bring-up and upstream fixes

- [Upstreamed initial NuttX support](#) for the Rockchip RK3588S SoC and Orange Pi 5 Pro, booting through U-Boot to NSH with GICv3, timer, UART, and PSCI support.
- [Fixed two EDID parsing bugs](#): incorrect standard-timing decoding and use of the wrong chromaticity macro.

Other upstream work

2026

- [Merged a heap-buffer-overflow fix](#) in NVIDIA `egl-wayland2` and [submitted a use-after-free fix](#) in `Wayland`.

Experience

Embedded Software Engineer Intern, Baykar Technologies

Feb 2025 – Jun 2025

T3 Gemstone platform

- Initiated the [Apache NuttX](#) port to the TI AM67A SoC: brought up an Arm Cortex-R5F core, configured the MPU and caches, defined the memory map, and implemented the early boot path.
- Wrote linker scripts, GPIO drivers, and UART drivers; debugged NuttX ELF loading through Linux `remoteproc` using `readelf` to inspect load addresses, sections, vectors, and resource-table requirements.

Avionics Software Developer, Atmaca Rocket Team

Sep 2022 – May 2023

University rocketry team · TÜBİTAK competition

- Developed redundant-controller avionics firmware and a parachute-deployment algorithm using Kalman-filtered multi-sensor altitude estimates; the avionics technical report received a full score from TÜBİTAK.

Education

B.Sc. Computer Science & Engineering, Akdeniz University

Sep 2022 – Jun 2026